

Table 1: Verification results for the experiments of HaliVer from Chapter 4.

Name	V	Result	Base			Qualifiers			Speedup
			#	T	σ_{mean}	#	T	σ_{mean}	
blur	0	✓	10	13	1	10	10	1	1.3
	1	✓	10	13	1	10	10	1	1.3
	2	✓	10	17	1	10	14	3	1.2
	3	✓	10	14	1	10	11	1	1.4
hist	0	✓	10	22	1	10	15	1	1.5
	1	✓	10	27	1	10	20	2	1.4
	2	✓	10	36	1	10	32	10	1.1
	3	✓	10	46	1	10	29	2	1.6
conv_layer	0	✓	10	144	2	10	60	6	2.4
	1	✓	10	162	1	10	65	1	2.5
	2	✓	10	220	3	10	74	5	3.0
	3	✓	10	201	2	10	72	5	2.8
gemm	0	✓	10	56	1	10	21	1	2.6
	1	✓	10	110	1	10	35	1	3.2
	2	✓	10	152	8	10	53	6	2.9
	3	Error	10			10			
auto_viz	0	✓	10	18	1	10	21	4	0.9
	1	✓	9	34	1	10	45	10	1.3
		✗	1	266	-	0			
	2	✓	10	36	1	10	39	10	0.9
	3	✓	10	34	1	10	28	1	1.2
bilateral_grid		✓	10	48	1	10	39	1	1.2
camera_pipe		✓	10	205	7	10	122	14	1.7
depthwise_separable_conv		✓	10	201	2	10	143	1	1.4
Total				1831	26		956	26	1.9

Table 2: Verification results for the experiments of HaliVer from Chapter 4.

Name	V	Result	Base			Qualifiers			Speedup
			#	T	σ_{mean}	#	T	σ_{mean}	
blur	0	✓	10	13	1	10	9	1	1.4
	1	✓	10	14	1	10	10	1	1.4
	2	✓	10	18	1	10	13	1	1.4
	3	✓	10	20	1	10	20	7	1.0
hist	0	✓	10	25	1	10	26	7	1.0
	1	✓	10	30	1	10	17	1	1.7
	2	✓	10	42	1	10	23	1	1.8
	3	✓	10	79	3	10	32	2	2.5
conv_layer	0	✓	10	153	1	10	53	1	2.9
	1	✓	10	171	1	10	72	6	2.4
	2	✓	10	246	3	10	71	1	3.5
	3	✓	10	227	2	10	68	1	3.3
gemm	0	✓	10	61	1	10	22	1	2.8
	1	✓	10	133	1	10	45	4	2.9
	2	✓	10	250	11	10	131	18	1.9
	3	Error	10			10			
auto_viz	0	✓	10	19	1	9	21	2	0.9
		✗	0			1	15	-	
	1	✓	8	63	1	10	52	1	2.1
		✗	2	290	1	0			
	2	✓	10	67	1	10	55	1	1.2
	3	✓	10	49	1	10	36	1	1.4
Total				1722	33		774	22	2.2

Table 3: Verification results for `step`, `sub_direction`, `solve_direction`, and `perform_iteration` produced by HaliVer. We use abbreviations for versions with concrete bounds (**CB**), nonconcrete bounds (**NCB**), **unique** and **const** type qualifiers, and no type qualifiers (**Normal**).

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	10	56	1	10	48	1	1.2
NCB	✓	10	56	1	10	50	1	1.1

(a) `step`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	10	466	9	10	142	4	3.3
NCB	✓	0			1	901	-	1.3
	✗	10	1028	58	9	780	89	

(b) `sub_direction`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	0			10	961	14	3.4
	✗	2	2086	1404	0			
	T.O.	8	-	-	0			
NCB	✗	9	1139	131	6	1211	468	
	T.O.	1	-	-	4	-	-	

(c) `solve_direction`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	0			10	2799	38	1.2
	✗	6	3146	101	0			
	T.O.	4	-	-	0			
NCB	✗	1	3589	-	0			
	T.O.	9	-	-	10	-	-	

(d) `perform_iteration`